9/07/24 Discrete Maths

(AUB) 1

Unit 1:- Sets

Set:- Collection of well defined objects is represented as set.

Ex:- No of flowers in the garden, No of bicycles in a parking area.

Types of sets:-

1) Roster form → All the elements of the set are seperated by commas (,) and enclosed between { }.

Ex:- Set of all numbers from 1 to 10.

$\therefore A = \{1, 2, 3, \ldots, 10\}$

2) Set-builder form → All the elements of the set have a common property.

Ex:- If the set has all the elements which are prime numbers.

$\therefore A = \{x \mid x \text{ is a prime numbers}\}$

3) Finite set → If the set contains finite number of elements.

(4) Infinite set → If the set of all the elements are not countable then it is a infinite set.

Ex:- A = {1, 2, 3, 4 .....}

(5) Empty set → consists of no elements.

Ex:- ϕ (or) { }

(6) Equal sets → Two sets are said to be equal if they have same elements.

Ex:- A = {1, 3, 2, 4} & B = {4, 3, 2, 1}

(7) Equivalent sets → Two sets are said to be equivalent if they have same number of elements.

Ex:- A = {1, 2, 3, 4} & B = {a, b, c, d}

n(A) = 4 n(B) = 4

(8) Power set → A set of every possible sub set.

Ex:- A = {1, 2, 3} The subsets → { }; {1}; {2}

{2,3}; {1,2,3}; {3}; {1,2}; {1,3}

∴ Power set is denoted by P(A) & P(A) is defined as P(A) = 8

* If the set 'A' has "n" elements then we have 2^n power set.

⑨ Subset → when all the elements of set A belongs to set B then $A \subseteq B$.
(A is strictly subset of B).

Σ^r
$2^3 = 8$

Ex:- $A = \{2, 6, 7, 8\}$

$B = \{1, 2, 3, 5, 7, 6, 8\}$

⑩ Universal set → Any set that contains all the sets under consideration.

30/07/24

Set Theory formula's

$n(A \cup B) = n(A) + n(B) - n(A \cap B)$

$n(A \cap B) = n(A) + n(B)$, where A & B are disjoint sets.

→ Disjoint sets:- A & B are said to be disjoint sets if there is no common element in two sets.

$A \cap B = \{\ \}$

$A = \{1, 2, 3\}$
$\cap$
$B = \{4, 5, 6\}$

→ $n(A^c) = n(\mu) - n(A)$

$\bar{A}$

$n(A-B) = n(A) - n(A\cap B)$

$n(A\cup B\cup C) = n(A) + n(B) + n(C) - n(A\cap B) - n(C\cap A) + n(A\cap B\cap C)$

Problems

① $U = \{1, 2, 3, 4, 5, 6, 7, 8\}$

$A = \{1, 2, 4\}$ $B = \{3, 4, 5, 6\}$ $C = \{3, 4, 7\}$

(i) $(A\cap B)\cup(A\cap C)$

(ii) $(A-B)^c$

(i) Sol:- $(A\cap B) = \{4\}$, $(A\cap C) = \{4\}$

$(A\cap B)\cup(A\cap C) = \{4\}\cup\{4\}$

$= \{4\}$

(ii) Sol:- $(A-B)$ = we shouldn't consider common elements.

$\{A-B\} = \{1, 2\}$

A^c

B^c

$(A-B)^c = U-(A-B)$

$= \{1, 2, 3, 4, 5, 6, 7, 8\} - \{1, 2\}$

$= \{3, 4, 5, 6, 7, 8\}$

2) In a group of 60 people 27 like cool drinks & 42 ~~drink~~ like hot drinks & each person likes atleast one of two drinks, identify the number of people both cold & hot drinks.

Sol:- Let 'C' be no of cold drinks $n(C) = 27$.

'H' be the no of hot drinks $n(H) = 42$

Total no of the people $n(C \cup H) = 60$

no of people like cold & hot drinks $= n(C \cap H)$

$$\Rightarrow n(C \cup H) = n(C) + n(H) - n(C \cap H)$$

$$\Rightarrow n(C \cap H) = n(C) + n(H) - n(C \cup H)$$

$$= 27 + 42 - 60$$

$$= 69 - 60$$

$$= 9$$

1) There are 35 students in arts class an 57 students in dance class. Find the no of stundents who are either in arts class or in dance class.

i) When two classes meet at different hours & 12 students are enrolled in both the activiti

ii) When two classes meet at same hour no student is enrolled in both activities.

* either, or - $\cup$ (union)
* and - $\cap$ (intersection)

Sol:- $n(A) = 35$

$n(B) = 57$.

$n(A \cup B) = ?$ $\quad n(A \cap B) = 12$

(i) $n(A \cup B) = n(A) + n(B) - n(A \cap B)$

$= 35 + 57 - 12$

$= 92 - 12$

$= 80$

(ii) $n(A \cap B) = 0$ no students are common.

$n(A \cup B) = n(A) + n(B) - n(A \cap B)$

4 Draw the venn diagrams.

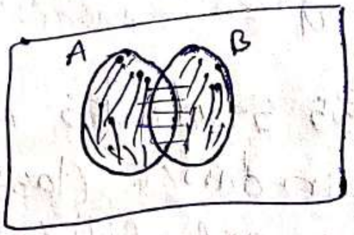
(i) $A \cup B$

(ii) $A \cap B$

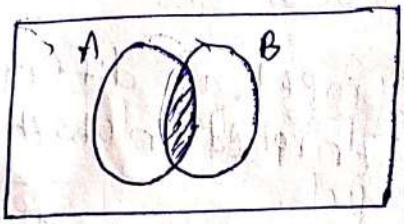
(iii) $A - B$

(iv) $B - A$.

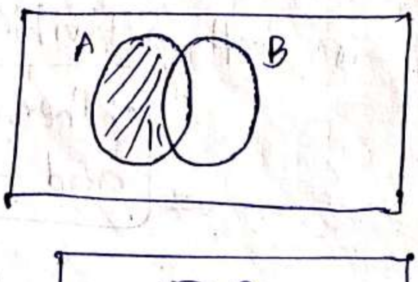
Sol:- $A \cup B \Rightarrow$



$A \cap B \Rightarrow$



$A - B. \Rightarrow$



$B - A \Rightarrow$

) There are 30 students in the class among them 8 students are learning both English & French. A total of 18 student learning English if every student is learning at least one language how many students are learning french in total?

Sol:-
$n(A) = 18$

$n(B)$ = French speaking

no of students learning in both English & French $= n(A \cap B) = 8$.

$n(A \cup B) = 30$.

$n(A \cup B) = n(A) + n(B) - n(A \cap B)$

$30 = 18 + n(B) - 8$

$30 = 10 + n(B)$

$n(B) = 20$.

) In a competition a school awarded medals in different categories, 36 medals in Dance, 12 medals in Dramatic and 18 medals in music. These medals went to a total of 45 person and only 4 persons got medals in all these categories. How many recieved medals in exactly two of these categories.

:- Let, 'A' be the no of medals in dance $n(A) = 36$

'B' be the no of medals in dramatics $n(B) = 12$

'C' be the no of medals in music $n(C) = 18$

Scanned with OKEN Scanner

no of medals in all the 3 categories

$= n(A \cap B \cap C) = 4$

Total no of medals $\Rightarrow n(A \cup B \cup C) = 45$

no. of medals recieved exactly two categories

= A & B or B & C or C & A

$= n(A \cap B) - n(A \cap B \cap C) + n(B \cap C) - n(A \cap B \cap C)$.

$n(C \cap A) - n(A \cap B \cap C)$

$= n(A \cap B) + n(B \cap C) + n(C \cap A) - 3 \times n(A \cap B \cap C)$

$= n(A \cap B) + n(B \cap C) + n(C \cap A)$.

$= 36 + 12 + 18 + 4 - 45$

$= 25$

→ no of medals recieved in 2 categories

$= 25 - 12 = 13$.

⑦ In a school all peoples play Hockey, Foot-ball ~~or Golf~~. 400 play foot ball, 150 play Hockey & 130 play both the games (i) Find the no of people who play football only.

(ii) The no of students ~~play Hockey~~.

Sol:- $n(A) = 150$

$n(B) = 400$

$n(A \cap B) = 130$

$n(A \cup B)$ = Total students.

$n(A - B)$ = no of students in Hockey.

$n(A) - n(A \cap B)$

$= 150 - 130$

$= 20$

(ii) no of students who play in foot ball only $= n(B - A)$

$= n(B) - n(A \cap B)$

$= 400 - 130$

$= 270.$

(iii) Total no of students $= n(A \cup B)$

$n(A \cup B) = n(A) + n(B) - n(A \cap B)$

$= 150 + 400 - 130$

$= 420.$

8) Draw the venn diagram for $(A - B) \cup (A - C)$ find the value

$A = \{1, 2, 4\}$; $B = \{3, 4, 5, 6\}$; $C = \{3, 4, 7\}$

$A - B = \{1, 2\}$

$A - C = \{1, 2\}$

$(A - B) \cup (A - C) = \{1, 2\}$

05/08/24 Session - 3

① Among a group of students 50 play cricket, 50 play Hockey and 40 play volleyball, 15 play both cricket and Hockey, 20 play both Hockey and volley ball, 15 play both cricket and volley ball and 10 play all the games.

(i) If every student play atleast one game find the number of students and how many play only cricket, only Hockey and only volley ball.

Sol:- Cricket $n(A) = 50$

Hockey $n(B) = 50$

Volley ball $n(C) = 40$

Cricket & Hockey $n(A \cap B) = 15$

Hockey & Volley ball $n(B \cap C) = 20$

Volley ball & Cricket $n(C \cap A) = 15$

All the 3 games $= n(A \cap B \cap C) = 10$

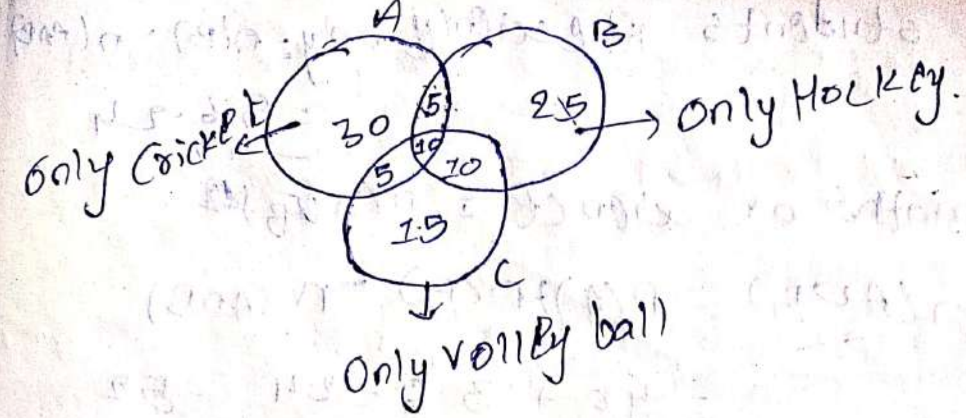
Total no of students $= n(A \cup B \cup C)$

$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C)$
$- n(C \cap A) + n(A \cap B \cap C)$.

$= 50 + 50 + 40 - 15 - 20 - 15 + 10$

$=$

Scanned with OKEN Scanner



Cricket only $= n(A) - 5 - 10 - 5$

$= 50 - 20$

$= 30$

Hockey Only $= n(B) - 5 - 10 - 10$

$= 50 - 25 = 25$

Volleyball only $= n(C) - 5 - 10 - 10$

$= 40 - 25 = 15$

2) In a class of 60 students, 40 students like maths, 36 like science, 24 like both the both subjects. (i) find the no of students who like maths only, (ii) Science only. (iii) Either Maths or science, (iv) Neither Maths nor science.

Sol:- (i) Like Maths $n(A) = 40$

Like science $n(B) = 36$

no of students like both maths & science $= n(A \cap B)$

$= 24$

Maths only $= n(A) - n(A \cap B)$

$= 40 - 24 = 16$

(ii) no of students like science only $= n(B) - n(A\cap B)$

$= 36 - 24$

$= 12$

(iii) Both maths or science $= n(A\cup B)$.

$$n(A\cup B) = n(A) + n(B) - n(A\cap B)$$

$$= 40 + 36 - 24 = 52$$

(iv) no of students like neither maths nor science $= n(A\cup B)^c$.

$$n(A\cup B)^c = n(U) - n(A\cup B)$$

$$= 60 - 52 = 8.$$

(3) Among the integers from 1 and 200. Find the number of integers that are divisible by either 2, 3 or 5.

Sol:- $A = \{n / n \le 200, 2 | 200\} = 100 \Rightarrow |A| = 100$

$B = \{n / n \le 200, 3 | 200\} = 66 \Rightarrow |B| = 66$

$C = \{n | n \le 200, 5 | 200\} = 40 \Rightarrow |C| = 40$

$A\cap B = \{n | n \le 200, 6 | 200\} = 33 \Rightarrow |A\cap B| = 33$

$B\cap C = \{n | n \le 200, 15 | 200\} = 13 \Rightarrow |B\cap C| = 13$

$C\cap A = \{n | n \le 200, 10 | 200\} = 20 = |C\cap A| = 20$

Scanned with OKEN Scanner

$A\cap B\cap C = \{n \mid n \le 200,\ 30 \mid 200\} = 6$ $|A\cap B\cap C| = 6$
$2\times 3\times 5$

no. from 1 to 200 divisible by either 2 or 3 or 5 $= |A\cup B\cup C|$

$|A\cup B\cup C| = |A| + |B| + |C| - |A\cap B| - |B\cap C| - |C\cap A| + |A\cap B\cap C|$

$= 100 + 66 + 40 - 33 - 13 - 20 + 6$

$= 146$

Inclusion & Exclusion Principle

* Let $|A|$ and $|B|$ indicates the cardinality of a set A & B. That is the number of elements of the set A & B. Then $A\cup B = |A| + |B| - |A\cap B|$

(4) Among the integers from 1 and 300. Find the number of integers that are divisible by either 2, 3, or 9.

$A = \{n \mid n \le 300;\ 2 \mid 300\} = 150$

$B = \{n \mid n \le 300;\ 3 \mid 300\} = 100$

$C = \{n \mid n \le 300;\ 9 \mid 300\} = 33$

$A\cap B = \{n \mid n \le 300,\ 6 \mid 300\} = 50$

$B\cap C = \{n \mid n \le 300,\ 9 \mid 300\} = 33$

$C\cap A = \{n \mid n \le 300, 18 \mid 300\} = 16$ $= |C\cap A| = 16$

$A\cap B\cap C = \{n \mid n \le 300, 18 \mid 300\} = 16$ $|A\cap B\cap C|$

$|A\cup B\cup C| = |A| + |B| + |C| - |A\cap B| - |B\cap C| - |C\cap A|$
$+ |A\cap B\cap C|$
$= 150 + 100 + 83 - 50 - 33 - 16 + 16 = 200$

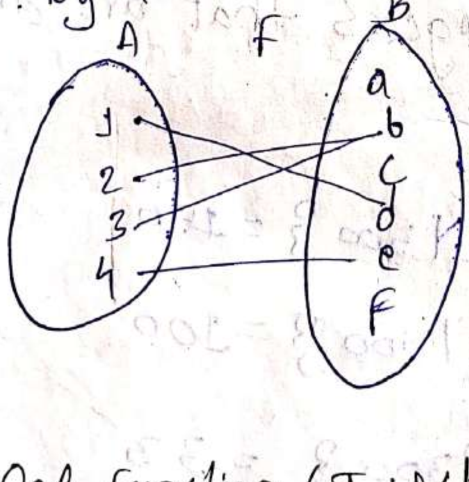
06/08/24

Session-4

Functions and Relations

* A function 'F' from a set A to a set B assigns each element of A to exactly one element of 'B'. And is denoted by $F: A \to B$ i.e; $f(a) = b$. to denote the element of 'A' assigns to the element of 'B'. By the function 'F'.

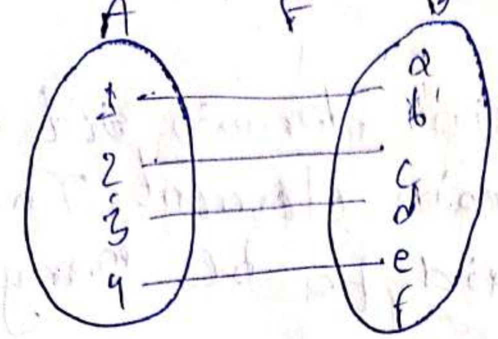
Ex:-



One-One Function (Injective)

→ A Function 'F' is said to be one to one if distinct elements of A mapped to distinct elements of 'B'. i.e; $f(a) = f(b) \Rightarrow a = b$

Ex:-



Onto Function (surjective)

* A function F from A to B ($F: A \rightarrow B$) is called an onto function if every element $b \in B$ there is an element $a \in A$ such that $f(a) =$ i.e; If the range of 'f' = codomain set

Into function

* If the range of f is subset of codomain set B i.e; $F(A) \subset B$. Then 'f' is said to be into function.

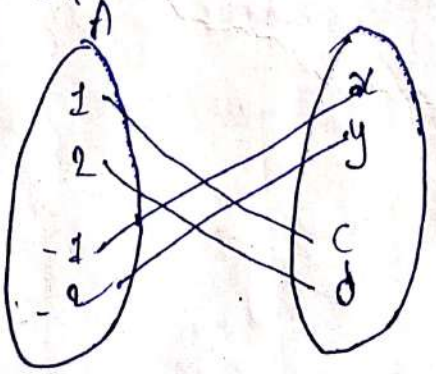
Range of $f = f(A) = \{b, c, d, e\}$

codomain, $B = \{a, b, c, d, e, f\} \Rightarrow F(A) \subset B$

Bijective Function

* A function 'F' is said to be both one-one and onto.

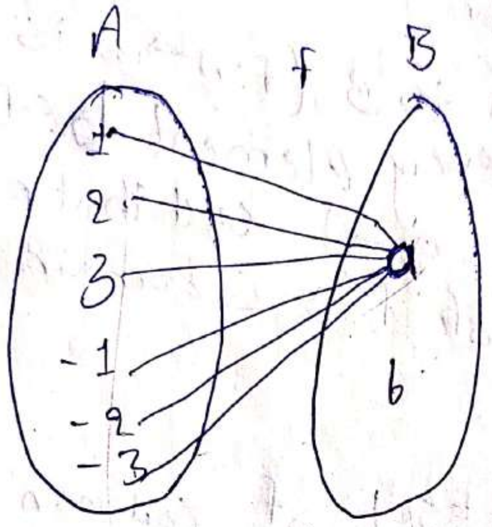
Ex:-



Many-one Function

*If all the elements in domain set maps to one and only codomain element. Then that function is said to be many one.

Ex-



Inverse Function

If $f : A \to B$ then $f^{-1} : B \to A$

If $x \in A$ & $y \in B$ ∋ $f(x) = y$

$x = f^{-1}(y)$

Q. Among the int

Session 12/08/24

RELATIONS

Def:- A set of ordered pairs defines a binary relation or relation.

* Relations represent relationship between elements of two sets.

* If R is a relation a particular ordered pair $x, y \in R$ can be written as xRy and

Ex: x is in Relation(R) to y.

Ex:- Represent the relation father to child in set-builder form.

$R = \{(x, y) / x \text{ is a father of } y\}$

Ex:- Define a relation between $A = \{5, 6, 7\}$ and $B = \{x, y\}$

$R: A \times B$

$= \{(5, x), (5, y), (6, x), (6, y), (7, x), (7, y)\}$

Inverse Relation

* If R is the relation from A to B $(R: A \times B)$ then $R^{-1}: B \to A$.

$R^{-1}: B \times A = \{(y, x) / y \in B, x \in A\}$ $R = \{(1, 2), (2, 3), (3, 4)\}$

Domain of R

- Set of all first elements of the ordered pairs in R.

Range of R.

* Set of all second elements in the ordered pairs of R.

Reflexive

If $a \in A$ then $(a,a) \in R$

Symmetric

If $(a,b) \in R$ the $(b,a) \in R$

Transitive

If $(a,b) \in R$ & $(b,c) \in R \Rightarrow (a,c) \in R$

Antisymmetric

If $(a,b) \in R$ & $(b,a) \in R$ then $a=b$

* A relation 'R' is said to be equivalence relation if 'R' is reflexive, symmetric & transitive.

* A relation 'R' is said to be partial order relation if 'R' is reflexive, transitive and antisymmetric.

Composition of Relations

If R is a relation for x to y & S is a relation for y to z

$R: x \to y$ & $S: y \to z$ $sof: x \to z$

then, $S \circ R : x \to z$

$= \{(x, z) \mid x \in x,\ z \in z\}$

$\Rightarrow R : A \times A$

$R^{C} : A \times A - R$

Q. Consider a relation $R = \{(1,1)\ (2,3)\ (3,4)\ (2,7)\}$ Find the domain and Range of R in terms of R^{-1}.

A) Domain of $R = \{1, 2, 3\}$

Range of $R = \{1, 3, 4, 7\}$

$R^{-1} = \{(1,1)\ (3,2)\ (4,3)\ (7,2)\}$

Domain of $R^{-1} = \{1, 3, 4, 7\}$ = Range of R

Range of $R^{-1} = \{1, 2, 3\}$ = Domain of R.

Q. Identify the properties of the following relation.

$R = \{(1,1)\ (1,2)\ (2,1)\ (2,2)\ (3,4)\ (4,1)\ (4,4)\}$

$A = \{1, 2, 3, 4\}$

Reflexive :- $a \in A$ then $(a,a) \in R$

$1 \in A \Rightarrow (1,1) \in R$

$2 \in A \Rightarrow (2,2) \in R$

$3 \in A \Rightarrow (3,3) \in R$

$\therefore$ R is not reflexive because $(3,3) \notin R$.

(ii) Sym.

Let R and S be the following relations on $A = \{1,2,3\}$, $R = \{(1,1)(1,2)(2,3)(3,1)(3,3)\}$

and $S = \{(1,2)(1,3)(2,1)(3,3)\}$

(i) $R \cap S$, $R \cup S$, R^c, $R \circ S$, $S \circ S$.

Sol:- (i) $R \cap S = \{(1,2)(3,3)\}$

(ii) $R \cup S = \{(1,1)(1,2)(2,3)(3,1)(3,3)(1,3)(2,1)\}$

(iii) $R^c = A \times A - R$

$= \{(1,1)(1,2)(1,3)(2,1)(2,2)(2,3)(3,1)(3,2)(3,3)\}$

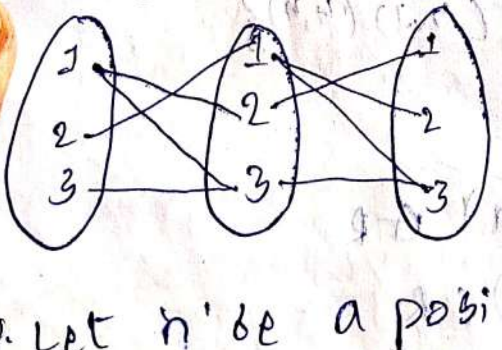
$- \{(1,1)(1,2)(2,3)(3,1)(3,3)\}$

$= \{(1,3)(2,1)(2,2)(3,2)\}$

(iv) $R \circ S = \{(1,3)(1,1),(2,1)(2,2)(3,1)(3,3)$
$(1,2),(2,3)\}$

(v) $S^2 = S \circ S$

S



$S \circ S = \{(1,1)(1,3)(2,2)(2,3)$
$(3,3)\}$

Q. Let n be a positive integer a relation R is defined on the set Z by a R b if and only if a-b is divisible by m for a,b belongs to z i.e; $R = \{(a,b) / a-b$ is divisible by $m\}$ for $a,b \in z$, $m \in z$ show that R is a equivalence relation.

Sol:- $R = \{(a,b) \mid |a-b|$ is divisible by $m\}$

(i) Reflexive

If $a \in Z$ then $|a-a| \div m$

$\Rightarrow 0 \div m$

$\therefore (a,a) \in R$.

$(a,a) \in R$

(ii) Symmetric

If $(a,b) \in R$ i.e; $|a-b| \div m \Rightarrow |b-a| \div m \Rightarrow (b,a) \in R$.

$\therefore R$ is symmetric.

(iii) Transitive

If $(a,b)\,(b,c) \in R$ i.e; $|a-b| \div m$ and $|b-c| \div m \Rightarrow |a-c|$ is divisible by 'm'.

$\therefore (a,c) \in R \Rightarrow R$ is transitive.

* The Relation 'R' is equivalence relation.

Representing relations using matrices.

Suppose 'R' is a relation from A to B where

$A = \{a_1, a_2, a_3, \ldots, a_m\}$

$B = \{b_1, b_2, b_3, \ldots, b_n\}$

The relation can be represented by the matrix

$M_R = [m_{ij}]$ where $m_{ij} = 1$ for $(a_i, b_j) \in R$

$= 0$ for $(a_i, b_j) \notin R$

Q. Let $A = \{1, 2, 3, 4\}$; $B = \{b_1, b_2, b_3\}$

$R = \{(1, b_2)\ (1, b_3)\ (3, b_2)\ (4, b_3)\}$

Determine the matrix of the relation.

Sol:-

A \ B	b_1	b_2	b_3
1	0	1	1
2	0	0	0
3	0	1	0
4	0	0	1

Q. Find the number of equivalence relations of the set $A = \{1, 2, 3, 4\}$

Sol:- To make bel tree write the last number of the previous row and add column wise.

$n = 1$ $\{1\} \Rightarrow 1$

$n = 2$ $\{1, 2\} \Rightarrow 1 \quad 2$

$n = 3$ $\{1, 2, 3\} = 2 \quad 3 \quad 5$

$n = 4$ $\{1, 2, 3, 4\} = 5 \quad 7 \quad 10 \quad 15 \rightarrow$ No of equivalence relations.

Q. Examine that the relation R is an equivalence relation in the $A = \{1, 2, 3, 4, 5\}$ given by the relation $R = \{(a, b) \mid a - b$ is even$\}$

Scanned with OKEN Scanner

Q. A set of integers, a relation 'R' is defined by xRy if and only if $x-y$ is divisible by '4'. Then verify 'R' is an equivalence relation.

(1) A.) $A = \{1, 2, 3, 4, 5\}$

$n = 1 \quad \{1\} \to 1$

$n = 2 \quad \{1, 2\} \Rightarrow 1 \quad 2$

$n = 3 \quad \{1, 2, 3\} = 2 \quad 3 \quad 5$

$\quad\quad 10 \quad 15$

$n = 4 \quad \{1, 2, 3, 4\} = 5 \quad 7$

$n = 5 \quad \{1, 2, 3, 4, 5\} = 15 \quad 20 \quad 27 \quad 37 \quad 52$

No of equivalence relations.

(2) A.) $R = \{(x-y) / x-y \text{ is divisible by } 4\}$

Reflexive:- If $a \in A$ then $(a, a) \in R$.

$(x, x) \in R \Rightarrow x - x = 0$ is divisible by 4.

'R' is reflexive.

Symmetric:- If $(a, b) \in R$ then $(b, a) \in R$

If $(x, y) \in R \Rightarrow (x - y)$ is divisible by 4

$\Rightarrow (y - x)$ is also divisible by 4

$\Rightarrow (y, x) \in R$

Transitive:- If $(a,b) \in R$ & $(b,c) \in R$ then $(a,c) \in R$

Suppose $(x,y) \in R$ & $(y,z) \in R$

$\Rightarrow (x-y)$ is divisible by 4 & $(y-z)$ is divisible by 4.

$\Rightarrow (x-z)$ is also divisible by 4 $(x,z) \in R$.

Session-5

13/08/24

Q. Show that the inclusion relation subset is partial ordering in the power set of a set 'S'.

Sol:- Let $P(S)$ = set of all subsets of 'S'.

$P(S) = \{\phi, A, B, C, D \ldots\}$

The inclusion relation is given by

$R = \{(A,B) \in P(S) \times P(S) / A \subseteq B\}$

R:- $(A,A) \in R$ $A \subseteq A$

$A \in P(S)$

Antisymmetric: $(A,B) \in R \Rightarrow A \subseteq B$

$(B,A) \in R \Rightarrow B \subseteq A$

$A = B$

Transitive:- $(A,B) \in R$, $(B,C) \in R$

$A \subseteq B$, $B \subseteq C$

$A \subseteq B \subseteq C$

$A \subseteq C$

Q. Verify that the relation less than or equal to is a partial ordering on Z. Find the relation

Sol:- $R = \{(a,b) \in Z \times Z / a \leq b\}$

Reflexive: $a \leq a \Rightarrow (a,a) \in R$

Anti symmetric:- $(a,b) \in R$ $(b,a) \in R$

i.e; $a \leq b$ & $b \leq a$

$\Rightarrow a = b$

Transitive

$(a,b) \in R$ & $(b,c) \in R$

$\Rightarrow a \leq b$ & $b \leq c$

$\Rightarrow a \leq c$

$\Rightarrow (a,c) \in R$

Q. Hasse diagram

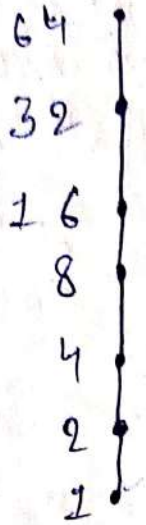
* It is a graphical representation of the relation of elements of a partially ordering set (poset). A point is drawn for each element for the partially ordered set and joined with the line segment with the following rules.

1) If $p < q$ then the point corresponding to p appears lower in the drawing than the point corresponding to q.

Two points p and q will be joined by line segment if 'p' is related to 'q'.

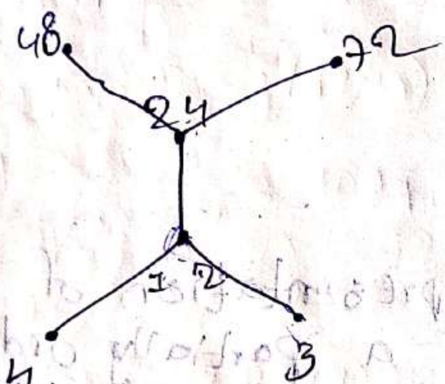
Q. Draw the Hasse diagram for $\{1,2,4,8,16,32,$

Sol:-



Q. Draw the Hasse diagram for $\{3,4,12,24,48,72\}$

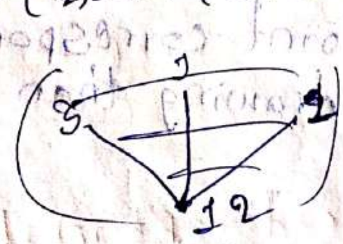
A = $\{(3,12)(3,24)(3,48)(3,72)(4,12)(4,24)(4,48)$
$(4,72)(12,24)(12,48)(12,72)(24,48)(24,72)\}$

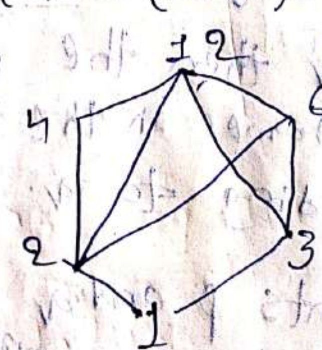


Q. Draw the Hasse diagram for $(D_{12}, /)$

$\{\cancel{\{3,4,12\}}$,

A) $R = \{(1,12)(2,12)(3,12)(4,12)(6,12)(12,12)\}$

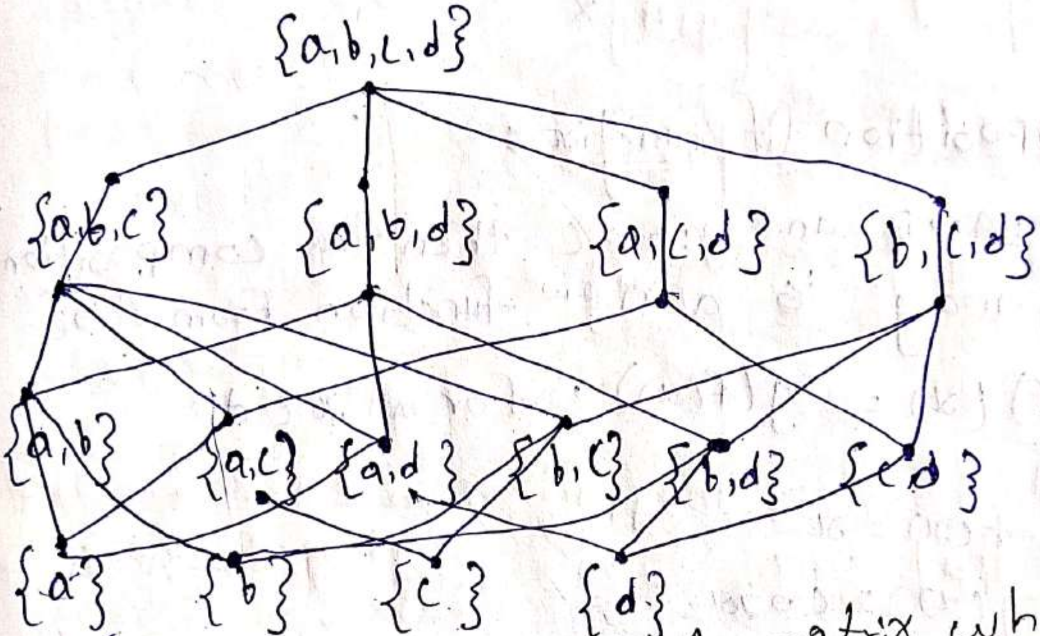




Q. Draw the Hasse diagram using the relation set inclusion on the set P(S) where

$S : \{a, b, c, d\}$

Sol:- $P(S) = \{\phi, \{a\}, \{b\}, \{c\}, \{d\}, \{a,b\}, \{a,c\}, \{a,d\}, \{b,c\}, \{b,d\}, \{c,d\}, \{a,b,c\}, \{a,b,d\}, \{a,c,d\}, \{b,c,d\}, \{a,b,c,d\}\}$



Q. Represent the relation into matrix where

$R = \{(1,1)\ (1,2)\ (2,3)\ (3,4)\ (2,1)\ (4,2)\ (4,3)\}$

Q. Find the no of equivalence relation for

$A = \{a, b, c, d, e\}$

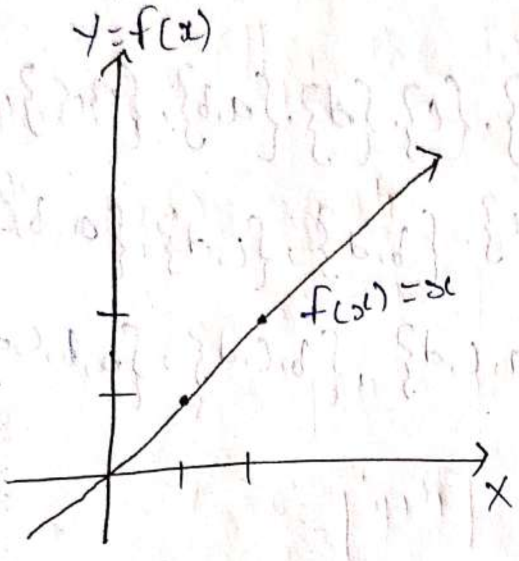
(1) A.)

A \ A	1	2	3	4
1	1	1	0	0
2	1	0	1	0
3	0	0	0	1
4	0	1	1	0

19/06/24

Identity function

The function $f: A \to A$ is called identity function if $f(x) = x$



$f(0) = 0$

$f(1) = 1$

$f(-2) = -2$

$f\left(\frac{1}{\sqrt{2}}\right) = \frac{1}{\sqrt{2}}$

Composition of functions

If $f: A \to B$ and $g: B \to C$ then the composition of 'f' and 'g' is a new function from $A \to C$

$(g \circ f)(x) = g(f(x))$, for all $x \in A$

Ex:- $f(x) = x^3$

$g(x) = \cos x$

$f \circ g(x) = f(g(x))$	$g \circ f(x) = g(f(x))$
$= f(\cos x)$	$= g(x^3)$
$= \cos^3 x$	$= \cos x^3$

$\therefore (f \circ g)(x) \neq (g \circ f)(x)$

Note:-

1. $g \circ f$ is defined, whereas $f \circ g$ need not be defined. So that, composition of function is not commutative
2. composition of function is associative,

that means if $f: A \to B$, $g: B \to C$ and $h: C \to D$ are three functions then $ho(gof) = (hog)of$

(3) If $f: A \to B$ and $g: B \to A$ then the function of 'g' is called the inverse of the function 'f' if $\boxed{gof = I_A \text{ and } fog = I_B}$.

Properties of Inverse function

→ The inverse of a function if exists is unique.

→ The necessary and sufficient condition for the function 'f' to be invertible is that f is one to one and onto.

→ If 'f' and 'g' are invertible functions then gof is also invertible and $(gof)^{-1} = f^{-1}og^{-1}$

Q. The formula for converting from farenheit to celsius temp is $y = \frac{5x}{9} - \frac{160}{9}$. Find the inverse of this function and determine whether the inverse is also a function or not.

Sol:-

$$y = \frac{5x}{9} - \frac{160}{9}$$

$$y = \frac{1}{9}[5x - 160]$$

$$9y = 5x - 160$$

$$5x = 9y + 160$$

$$x = \frac{1}{5}[9y + 160]$$

$$f^{-1}(y) = \frac{1}{5}[9y + 160]$$

$$f^{-1}(x) = \frac{1}{5}[9x + 160]$$

$$f^{-1}(x) = \frac{9}{5}x + 32.$$

∴ Inverse is also a function.

Q. A simple cipher takes a number and quotes it, using the function $f(x) = 3x-4$. Find the inverse of this function, determine whether the inverse is also a function.

Sol:-

$$f(x) = 3x-4$$

$$y = 3x-4$$

$$3x = y+4$$

$$x = \frac{1}{3}[y+4]$$

$$f^{-1}(y) = \frac{1}{3}[y+4]$$

$$f^{-1}[x] = \frac{1}{3}[x+4]$$

∴ Inverse is also a function.

Q. Let f is a mapping from R to R $f: R \to R$ such that $f(x) = x^2+1$ find $f^{-1}(10) = ?$

Sol:-

$$y = x^2+1$$

$$y-1 = x^2$$

$$x = \sqrt{y-1}$$

$$f^{-1}(y) = \sqrt{y-1}$$

$$f^{-1}(x) = \sqrt{x-1}$$

$$f^{-1}(10) = \sqrt{10-1}$$

$$= \sqrt{9}$$

$$\therefore f^{-1}(10) = \pm 3$$

Q. Let $f: R \to R$ such that $f(x) = 4x+3 \; \forall x \in R$. Show that f is invertible and find f^{-1}.

Sol:- $f(x) = 4x+3$

$y = 4x+3$

$y-3 = 4x$

$x = \frac{y-3}{4}$

$$\boxed{f^{-1}(y) = \frac{y-3}{4}}$$

f is onto.

Check one-one or not.

$f(x_1) = 4x_1+3$

$f(x_2) = 4x_2+3$

$f(x_1) = f(x_2)$

$\cancel{4}x_1 + \cancel{3} = \cancel{4}x_2 + \cancel{3}$

$x_1 = x_2$

$\therefore$ It is one-one.

Check onto or not

$$\boxed{f^{-1}(y) = \frac{y-3}{4}} \to f \text{ is onto}$$

$\therefore$ f is one-one $\xrightarrow{}$ injective and onto $\xrightarrow{}$ surjective.

f is bijective function.

f is invertible

$f^{-1}(x) = \frac{x-3}{4}$

Q. Let $A = \{2, 3, 4, 5\}$ and $B = \{7, 9, 11, 13\}$. Let f is $f = \{(2,7), (3,9), (4,11), (5,13)\}$. Show that f is invertible and find f^{-1}.

Q. If $f(x) = x^2$ & $g(x) = 3x$ and $h(x) = x-2$. Prove that $(fog)oh = fo(goh)$

$f(g(x)) \circ h$

$= (f(3x)) \circ h$

$((3x)^2) \circ h$

$(9x^2) \circ (x-2)$

$= 9 \cdot (x-2)^2$

$= 9(x^2+4-4x)$

RHS $= f \circ (g \circ h)$

$= f \circ (g(h(x)))$

$= f \circ g(x-2)$

$= f \circ 3(x-2)$

$= f \circ (3x-6)$

$= x^2 \circ (3x-6) = (3x-6)^2$

$= 9x^2+36-36x$

$\therefore (f \circ g) \circ h = f \circ (g \circ h)$

26/08/24

Session-7

$A = \{1,2,3,4\}$ and $B = N$ and $f: A \to B$ defined by $f(x) = x^3$ then find

(i) The range of 'f' and identify the type of function

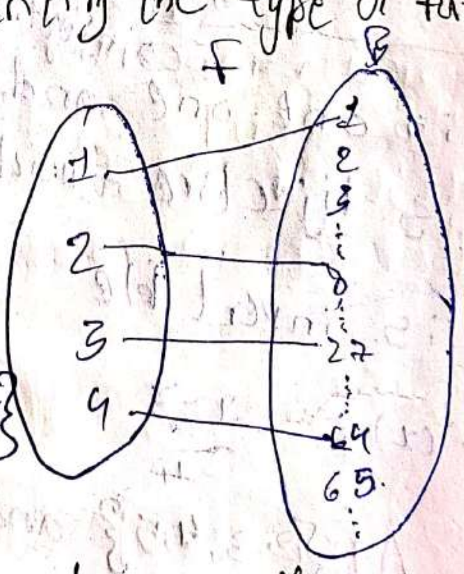
A) $f(1) = 1^3 = 1$

$f(2) = 2^3 = 8$

$f(3) = 3^3 = 27$

$f(4) = 4^3 = 64$

Range of $f = \{1, 8, 27, 64\}$



(ii) All the distinct elements in the domain set maps to distinct elements in the co-domain set. F is one-one.

(2) Consider the function $f(x) = 3x+2$. Verify whether it is bijective or not. If so, find it's inverse.

Sol:- If 'f' is a bijection then 'f' is said to be one-one and onto.

→ 'f' is one-one : If $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$

let $f(x_1) = f(x_2)$

$$3x_1 + \cancel{2} = 3x_2 + \cancel{2}$$

$$\cancel{3}x_1 = \cancel{3}x_2$$

$$\boxed{x_1 = x_2}$$ ∴ f is one-one.

→ 'f' is onto:- $f : R \to R$
$x \quad y$

$f(x) = y$

$3x + 2 = y$

$x = \frac{y-2}{3} \in R$ ∴ 'f' is onto.

∴ 'f' is said to be bijective function; it is also a inverse function.

$f(x) = y$

$x = f^{-1}(y)$

$\frac{y-2}{3} = f^{-1}(y)$

$$\boxed{\frac{x-2}{3} = f^{-1}(x)}$$

→ $f(x) = e^{x^2}$

$f(x_1) = f(x_2)$

$e^{x_1^2} = e^{x_2^2} \quad (x_1 \neq x_2)$

$x_1 = 2 \Rightarrow f(x_1) = e^{x_1^2} = e^{2^2} = e^4$

$x_2 = -2 \Rightarrow f(x_2) = e^{x_2^2} = e^{(-2)^2} = e^4$

$\therefore$ 'F' is not one-one function.

③ How many onto functions are there from an 'n' element set to a 2 element set.

→ Consider the no of elements in 'm' and 'n' $\therefore$ The number of possible onto function are $n^m - {}^nC_1 [n-1]^m + {}^nC_2 (n-2)^m + \ldots$

Sol:-

$m = n \; ; \; n = 2$

$2^n - 2C_1 [2-1]^n + 2C_2 (2-2)^n$

$2^n - 2 (1)^n + 1 (0)^n$

$= 2^n - 2(1)^n$ //.

④ There are real numbers 'a' and 'b' for which the function 'f' has the properties $f(x) = ax + b$. For all real numbers 'x'. Then $f(bx+a) = x$. what is the value of $a+b$.

Sol:- $f(x) = ax+b \quad f(bx+a) = x$

we have $F[F^{-1}(x)] = x$

Given $f(bx+a) = x$

$bx+a = f^{-1}(x)$

→ $F[f^{-1}(x)] = x$

$f[bx+a] = x$

$a[bx+a] + b = x$

$abx + a^2 + b = x$

coeff of $x \Rightarrow ab = 1$ $\quad$ $a(-a^2) = 1$

const $\Rightarrow a^2 + b = 0$ $\quad$ $a = -\frac{1}{a^2}$

$b = -a^2$ $\quad$ $a + b = +\frac{1}{a^2} - \frac{a^2}{1}$

session :8

(1) A function $f(x) = 3x - 4$. Find the inverse of this function

$f(x) = y$

$x = f^{-1}(y)$

$f(x) = 3x - 4$

$y = 3x - 4$

$x = \frac{y+4}{3}$

$f^{-1}(y) = \frac{y+4}{3}$

$\therefore \boxed{f^{-1}(x) = \frac{x+4}{3}}$

(2) If $A = \{1, 2, 3, 4, 5\}$ $B = \{1, 2, 3, 8, 9\}$ and the functions $f: A \to B$ & $g: A \to A$ defined by

$f = \{(1,8)\ (3,9)\ (4,3)\ (2,1)\ (5,2)\}$

$g = \{(1,2)\ (3,1)\ (2,2)\ (4,3)\ (5,2)\}$ then find

(i) fog (ii) gof (iii) fof (iv) gog if they exist.

(i) A) $fog(x) = f(g(x))$

$x = 1$ $\quad$ $= f(g(1))$

$= f(2) = \cancel{8} 1$

$x = 2 \Rightarrow f(g(2))$
$= f(2)$
$= 1$

$x = 3 \Rightarrow f(g(3))$
$= f(7)$
$= 8$

$x = 4 \Rightarrow f(g(4))$
$f(3)$
$= 9$

$x = 5 \Rightarrow f(g(5))$
$= f(2)$
$= 1.$

(ii) $gof = gof(x) = g(f(x))$

$x = 1 \quad = g(f(1))$
$= g(8) \times$ does not exist

$x = 2 = g(f(2))$
$= g(1)$
$= 2$

$x = 3 = g(f(3))$
$= g(9) \times$ does not exist.

(iii) $gog = gog(x) = g(g(x))$
$= g(g(1))$
$= g(2) = 2$

$x = 2 \Rightarrow g(g(2))$
$= g(2)$
$= 2$

$x = 3 \Rightarrow g(g(3))$
$= g(1)$
$= 2$

$x = 4 \Rightarrow g(g(4))$
$= g(3)$
$= 1.$

$x = 5 \Rightarrow g(g(5))$
$= g(2)$
$= 2.$

(iv) fof does not exist.

(3) If S is set of all elements $S=\{1,2,3,4,5\}$ if the function $f,g,h : S \to S$ are given by

$f=\{(1,2)(2,1)(3,4)(4,5)(5,3)\}$ &

$g=\{(1,3)(2,5)(3,1)(4,2)(5,4)\}$

$h=\{(1,2)(2,2)(3,4)(4,3)(5,1)\}$ inverse but explain why 'f' and 'g' has inverse but not 'h'.

A) $f(a)=b \Rightarrow a=f^{-1}(b)$

$f=\{(a,b)\}$ $f^{-1}=\{(b,a)\}$

$g=\{(a,b)\}$ $g^{-1}=\{(b,a)\}$

$f^{-1}=\{(2,1)(1,2)(4,3)(5,4)(3,5)\}$

$g^{-1}=\{(3,1)(5,2)(1,3)(2,4)(4,5)\}$

→ All the elements are not mapped in the domain set and the element '2' in the domain set mapped two times in the co-domain set. ∴ h^{-1} is not exist.

(4) $f(x)=x+2$; $g(x)=2x+1$

$fog(x)= f(g(x))$
$= f(2x+1)$
$= 2x+1+2 = 2x+3$

$gof(x)= g(f(x))$
$= g(x+2)$
$= 2(x+2)+1$
$= 2x+4+1$

(5) Show that $f(x) = 3x - 5$ is bijective function from $f: R \to R$

$f(x_1) = f(x_2)$

$3x_1 - \cancel{5} = 3x_2 - \cancel{5}$

$x_1 = x_2$.

'f' is one-one

$f(x) = y$

$x = f^{-1}(y)$

$y = 3x - 5$

$x = \frac{y+5}{3}$

$f^{-1}(y) = \frac{y+5}{3}$

$f^{-1}(x) = \frac{x+5}{3} \in R$

'f' is onto.

$\therefore$ 'f' is a bijective.

→ 'Floor function' is the function that takes as input a real number 'x' and gives as output the greatest integer $\leq x$

$\lfloor x \rfloor$

→ The 'ceiling function' maps x to the least integer $\geq x$ and is denoted by

$\lceil x \rceil$

Ex:- C.F of $\lceil 29.5 \rceil$
$= 30$

C.F of $\lceil -10.7 \rceil$
$= -10$

C.F of $\lceil -0.6 \rceil$
$= 0$

C.F of $\lceil 3.14 \rceil$
$= 4$

C.F of $\lceil 5\pi \rceil = \lceil 15.7 \rceil = 16$

C.F of $\lceil -3.14 \rceil = -3$

27/08/24

→ F.F of $\lfloor 3/2 \rfloor \to \lfloor 1.5 \rfloor = 1$

→ F.F of $\lfloor -4/3 \rfloor = \lfloor -1.3 \rfloor = -2$

→ F.F of $\lfloor 0.8 \rfloor = 0$

Boolean Function

→ It's a function that takes one or more binary maps (values that are either zero or one and representing false or true respectively and produce outputs that are also zero's & one).

$B = \{0, 1\}$ $\quad$ $B_n = \{x_1, x_2, x_3, \ldots, x_n\}$

$f : B_n \to \{0, 1\}$ $\quad$ $x, y \in B_n$

Compliment

$\bar{1} = 0$

$\bar{0} = 1$

x	y	$x+y$
1	1	1
1	0	1
0	1	1
0	0	0

(Sum +/OR)

$\boxed{\begin{array}{l} +/\text{OR } (\vee) \\ \cdot/\text{AND } (\wedge) \end{array}}$

x	y	$x.y$
1	1	1
1	0	0
0	1	0
0	0	0

Product (·/AND)

Q. Write the boolean function $f(x \vee y) = xy + \bar{y}$

Sol:-

$\wedge \to$ AND
$\vee \to$ OR

x	y	xy	$\bar{y}$	$xy+\bar{y}$
1	1	1	0	1
1	0	0	1	1
0	1	0	0	0
0	0	0	1	1

Q. Find the values of the boolean function represented by $f(x, y, z)$, $xy + \bar{z}$

x	y	z	xy	$\bar{z}$	$xy+\bar{z}$
1	1	1	1	0	1
1	1	0	1	1	0
1	0	1	0	0	0
0	1		0	0	1
1	0	0	0	1	1
0	1	0	0	1	0
0	0		0	0	
0	0	0	0	1	